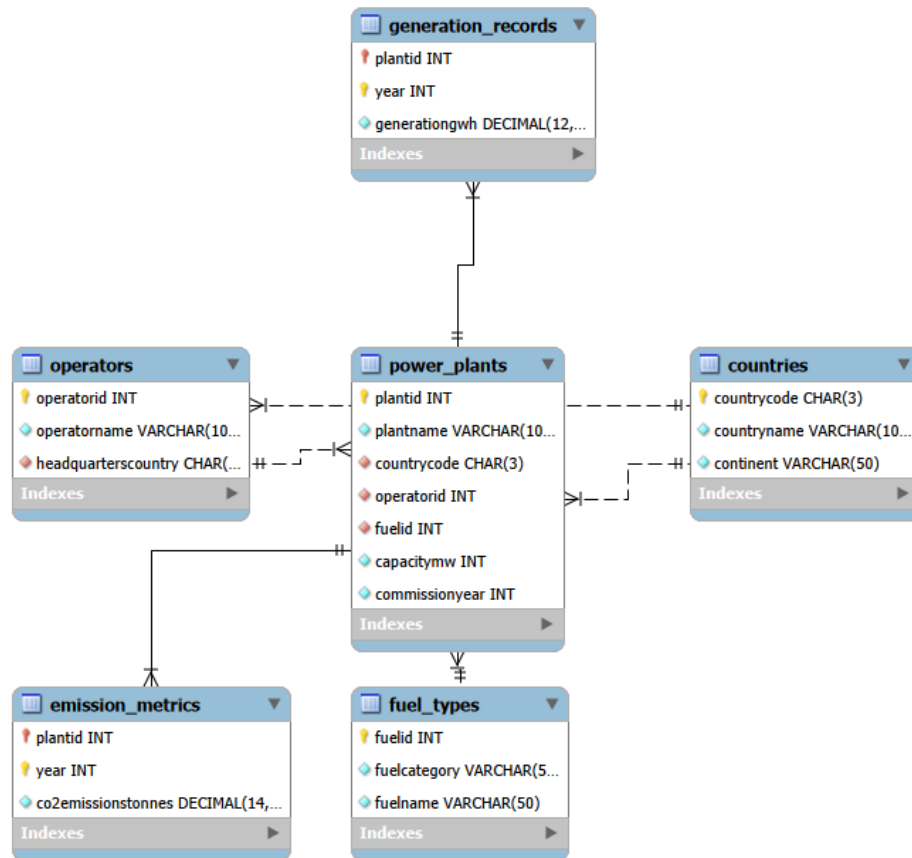


Volodymyr Siedykh
Assignment 3
Generated ER



SQL for each problem

- 1) Retrieve the plant name, country name, operator name,
- fuel category, fuel name, capacity (in MW), and commission year
- for all power plants without using table aliases. Sort the results
- in descending order by capacity.

```
select power_plants.plantname,  
       countries.countryname,  
       operators.operatorname,  
       fuel_types.fuelcategory,  
       fuel_types.fuelname,  
       power_plants.capacitymw,
```

```

power_plants.commissionyear
from power_plants
join countries using (countrycode)
join operators using (operatorid)
join fuel_types using (fuelid)
order by power_plants.capacitymw desc;

```

	plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
►	Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
	Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
	Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
	Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
	Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
	Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
	Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
	Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

-- 2) Retrieve the plant name, country code, calendar year, and annual generation (in GWh)
-- for all power plants for the year 2024 without using table aliases. Sort the results in
descending order by generation.

```

select plantname,
        countrycode,
        generation_records.year,
        generation_records.generationgwh
from power_plants
join generation_records using (plantid)
where generation_records.year = 2024
order by generation_records.generationgwh desc;

```

	plantname	countrycode	year	generationgwh	--
►	Three Gorges	CHN	2024	98800.00	
	Itaipu	BRA	2024	89000.10	
	Tuoketuo	CHN	2024	71000.30	
	Vindhyachal	IND	2024	35600.80	
	Grand Coulee	USA	2024	21500.50	
	Janschwalde	DEU	2024	18400.20	
	Loy Yang	AUS	2024	14200.40	
	Kashiwazaki-Kariwa	JPN	2024	0.00	

3) Retrieve the plant name, country code, calendar year, annual generation (in GWh),
-- and CO2 emissions (in tonnes) for all power plants for
-- the year 2024 without using table aliases. Sort the results in ascending order by CO2
emissions.

```

select plantname,
       countrycode,
       generation_records.year,
       generation_records.generationgwh,
       emission_metrics.co2emissionstonnes
from power_plants
join generation_records on power_plants.plantid = generation_records.plantid
join emission_metrics
on generation_records.plantid = emission_metrics.plantid
and generation_records.year = emission_metrics.year
where generation_records.year = 2024
order by emission_metrics.co2emissionstonnes;

```

	plantname	countrycode	year	generationgwh	co2emissionstonnes
►	Grand Coulee	USA	2024	21500.50	0.00
	Itaipu	BRA	2024	89000.10	0.00
	Kashiwazaki-Kariwa	JPN	2024	0.00	0.00
	Three Gorges	CHN	2024	98800.00	0.00
	Loy Yang	AUS	2024	14200.40	1950000.00
	Janschwalde	DEU	2024	18400.20	2450000.00
	Vindhyachal	IND	2024	35600.80	3100000.00
	Tuoketuo	CHN	2024	71000.30	5200000.00

```

-- 4) Write a SQL query using a Common Table Expression (CTE) to calculate
-- the total cumulative power generation (in GWh) for each operator across
-- all available years without using table aliases.
-- Retrieve the operator name, headquarters country, and their total generated power, and
-- sort the results in descending order by total generation.

```

```

with operator_generation as (
    select power_plants.operatorid,
           sum(generation_records.generationgwh) as totalgenerationgwh
    from power_plants
    join generation_records using (plantid)
    group by power_plants.operatorid
)
select operators.operatorname,
       operators.headquarterscountry,
       operator_generation.totalgenerationgwh
from operators
join operator_generation using (operatorid)

```

order by operator_generation.totalgenerationgwh desc;

	operatorname	headquarterscountry	totalgenerationgwh
►	China Yangtze Power	CHN	169800.30
	Electrobras	BRA	89000.10
	NTPC Limited	IND	35600.80
	Global Hydro Inc	USA	21500.50
	Rhein Power AG	DEU	18400.20
	Snowy Hydro	AUS	14200.40
	Tepeco Energy	JPN	0.00

-- 5) Write a SQL query using two separate Common Table Expressions (CTEs)
 -- one to calculate total power generation by country code and
 -- another to calculate total CO2 emissions by country code without using table aliases. Then,
 -- join these CTEs with the countries table to display the country name, total generation (in
 GWh),
 -- and total CO2 emissions (in tonnes). Sort the results in descending order by total generation.

```
with country_generation as (
  select power_plants.countrycode,
         sum(generation_records.generationgwh) as totalgenerationgwh
  from power_plants
  join generation_records using (plantid)
  group by power_plants.countrycode
),
country_emissions as (
  select power_plants.countrycode,
         sum(emission_metrics.co2emissionstonnes) as totalco2emissionstonnes
  from power_plants
  join emission_metrics using (plantid)
  group by power_plants.countrycode
)
select countries.countryname,
       country_generation.totalgenerationgwh,
       country_emissions.totalco2emissionstonnes
FROM countries
JOIN country_generation using (countrycode)
JOIN country_emissions using (countrycode)
ORDER BY country_generation.totalgenerationgwh DESC;
```

	countryname	totalgenerationgwh	totalco2emissionstonnes
►	China	169800.30	5200000.00
	Brazil	89000.10	0.00
	India	35600.80	3100000.00
	United States	21500.50	0.00
	Germany	18400.20	2450000.00
	Australia	14200.40	1950000.00
	Japan	0.00	0.00